

The primal simplex method

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We consider a linear program in standard form:

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \quad (\text{P}) \\ & x \geq 0_n, \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$. We assume $\text{rank}(A) = m$. The set $N := \{1, \dots, n\}$ is used to index the variables. The j^{th} column of A is denoted A_j . The dual linear program (D) of (P) is given by:

$$\begin{aligned} \max \quad & b^T y \\ \text{s.t.} \quad & A^T y \leq c, \quad (\text{D}) \end{aligned}$$

and the polyhedron:

$$F_P := \{x \in \mathbb{R}^n : Ax = b \text{ and } x \geq 0_n\}$$

denotes the set of feasible solutions to (P).

1 Bases and basis matrices

A *basis* of A is an m -subset $B \subseteq N$ such that the columns $\{A_i : i \in B\}$ are linearly independent. Two bases of A are *adjacent* if they only differ by one element.

Given an ordering of the basic columns one has a *basis matrix*. Specifically, given a fixed ordering $\langle i_1, \dots, i_m \rangle$ of the indices in $B = \{i_1, \dots, i_m\}$, the associated basis matrix is $A_B = [A_{i_1}, \dots, A_{i_m}]$. A basis matrix A_B is non-singular since the columns $\{A_i : i \in B\}$ are linearly independent.

The matrix $\bar{A} := A_B^{-1}A$ is called the *updated* A matrix. The j^{th} column of $\bar{A}$ is denoted $\bar{A}_j$. As argued in the note on the dual simplex method, the non-zero entries of $\bar{A}$ characterize the adjacent bases of B :

Observation 1 (*Characterization of adjacent bases of A*)

Let $B' := (B \cup \{j\}) \setminus \{i_k\}$ be the set obtained by replacing the basic column $i_k \in B$ with the non-basic column $j \in N \setminus B$. The set B' is a basis of $A \iff \bar{a}_{k,j} \neq 0$. ■

2 Basic solutions

The *primal basic solution* associated with B is the unique solution $\bar{x} \in \mathbb{R}^n$ to the system:

$$Ax = b \quad \text{and} \quad x_j = 0 \quad \text{for } j \in N \setminus B,$$

and the *dual basic solution* associated with B is the unique solution $\bar{y} \in \mathbb{R}^m$ to:

$$A_i^T \bar{y} = c_i \quad \text{for all } i \in B.$$

The primal basic solution $\bar{x}$ associated with B has $\bar{x}_j = 0$ on non-basic coordinates $j \in N \setminus B$, and the basic coordinates are obtained by solving the remaining system with m variables and m equalities:

$$A_B \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = b \iff \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = A_B^{-1} b.$$

A formula can also be given for the dual basic solution associated with B . Let $c^B \in \mathbb{R}^m$ be the vector of cost coefficients associated with basic variables only:

$$c^B := \begin{pmatrix} c_{i_1} \\ \vdots \\ c_{i_m} \end{pmatrix}.$$

The dual basic solution associated with B can then be calculated by solving a system of m equalities and m variables:

$$A_i^T \bar{y} = c_i \text{ for } i \in B \iff \bar{y}^T A_B = (c^B)^T \iff \bar{y}^T = (c^B)^T A_B^{-1}.$$

If the dual basic solution $\bar{y}$ associated with B satisfies $A_j^T \bar{y} \leq c_j$ for *all* $j \in N$, then $\bar{y}$ is called *basic feasible* for (D), and if the primal basic solution $\bar{x}$ satisfies $\bar{x}_j \geq 0$ for *all* $j \in N$, then $\bar{x}$ is called *basic feasible* for (P).

The primal and dual basic solutions $\bar{x}$ and $\bar{y}$ (not necessarily feasible) have the same objective function value in their respective problems:

$$\bar{y}^T b = (c^B)^T A_B^{-1} b = (c^B)^T \begin{pmatrix} \bar{x}_{i_1} \\ \vdots \\ \bar{x}_{i_m} \end{pmatrix} = c^T \bar{x}.$$

From weak duality it then follows that, if $\bar{x}$ is primal feasible, and if $\bar{y}$ is dual feasible, then $\bar{x}$ is optimal for (P), and $\bar{y}$ is optimal for (D).

3 A translated cone associated with every basis

We now consider a polyhedron associated with a basis B :

$$F_P(B) := \{x \in \mathbb{R}^n : Ax = b \text{ and } x_j \geq 0 \text{ for } j \in J\},$$

where $J := N \setminus B$ is used to index the non-basic variables for simplicity. The polyhedron $F_P(B)$ is obtained from F_P by eliminating the non-negativity constraints on the basic variables. Hence $F_P(B)$ is a relaxation of F_P , *i.e.*, one has $F_P \subseteq F_P(B)$. The polyhedron $F_P(B)$ gives relevant "local" information at $\bar{x}$: All inequalities that are not tight at $\bar{x}$ are removed from the description.

The polyhedron $F_P(B)$ has exactly one basic feasible solution, namely the primal basic solution $\bar{x}$ obtained by setting *all* inequalities of $F_P(B)$ to equality. Hence $F_P(B)$ is a translated cone, *i.e.*, one has $F_P(B) = \bar{x} + \text{rec}(F_P(B))$, where $\text{rec}(F_P(B))$ is the recession cone of $F_P(B)$:

$$\text{rec}(F_P(B)) = \{x \in \mathbb{R}^n : Ax = 0_m \text{ and } x_j \geq 0 \text{ for } j \in J\}$$

We now characterize the extreme directions of $\text{rec}(F_P(B))$. We will show that the extreme rays of $\text{rec}(F_P(B))$ are in one-to-one correspondence with the extreme points of the polyhedron:

$$F^* := \{x \in \text{rec}(F_P(B)) : \sum_{j \in J} x_j = 1\}$$

The polyhedron F^* is obtained from $\text{rec}(F_P(B))$ by intersecting $\text{rec}(F_P(B))$ with the hyperplane $\sum_{j \in J} x_j = 1$.

Observe that the recession cone of F^* simply consists of the origin:

$$\begin{aligned} \text{rec}(F^*) &= \{x \in \mathbb{R}^n : Ax = 0_m, \sum_{j \in J} x_j = 0 \text{ and } x_j \geq 0 \text{ for } j \in J\} \\ &= \{x \in \mathbb{R}^n : Ax = 0_m \text{ and } x_j = 0 \text{ for } j \in J\} = \{0_n\}, \end{aligned}$$

where the last equality holds since B is a basis of A .

Hence F^* is the convex hull of its extreme points (=basic feasible solutions). We now argue that the extreme points of F^* suffice to describe the recession cone $\text{rec}(F_P(B))$ of $F_P(B)$:

Lemma 1 (*Description of the extreme directions of $\text{rec}(F_P(B))$*)

Let $X \subseteq F^*$ denote the extreme points of F^* . One has $\text{rec}(F_P(B)) = \text{Cone}(X)$. ■

Proof: " $\supseteq$ ": One has $F^* \subseteq \text{rec}(F_P(B))$ since F^* is obtained from $\text{rec}(F_P(B))$ by adding an extra equality. Hence $X \subseteq F^* \subseteq \text{rec}(F_P(B))$. Applying $\text{Cone}(\cdot)$ on both sides of the inclusion $X \subseteq \text{rec}(F_P(B))$ gives $\text{Cone}(X) \subseteq \text{rec}(F_P(B))$.

" $\subseteq$ ": Now let $\bar{x} \in \text{rec}(F_P(B))$ be arbitrary. We must show $\bar{x} \in \text{Cone}(X)$. Scaling $\bar{x}$ gives $\bar{y} := \frac{1}{\sum_{j \in J} \bar{x}_j} \cdot \bar{x} \in F^* = \text{Conv}(X)$. Scaling $\bar{y}$ then gives $(\sum_{j \in J} \bar{x}_j) \cdot \bar{y} = \bar{x} \in \text{Cone}(\text{Conv}(X)) = \text{Cone}(X)$. ■

We now provide formulas for the extreme directions of $\text{rec}(F_P(B))$ (=extreme points of F^*). Let $\bar{x}$ be an arbitrary extreme point of F^* . One cannot have that $\bar{x}$ satisfies *all* inequalities $x_j \geq 0$ for $j \in J$ with equality, since this implies $\bar{x} = 0_n$ which contradicts $\sum_{j \in J} \bar{x}_j = 1$.

The point $\bar{x}$ is a basic feasible solution for F^* . Hence $\bar{x}$ is satisfied with equality by at least n of the inequalities/equalities defining F^* . Since F^* is described by $n+1$ inequalities and equalities, this implies that $\bar{x}$ satisfies exactly $|J| - 1$ of the inequalities $x_j \geq 0$ for $j \in J$ with equality. We conclude that $\bar{x}$ is a solution to:

$$Ax = 0_m, \quad x_j = 1 \quad \text{and} \quad x_k = 0 \quad \text{for } k \in J \setminus \{j\} \quad (1)$$

for some $j \in J$.

Lemma 2 (*Description of the polyhedron $F_P(B)$*)
The polyhedron $F_P(B)$ is given by:

$$F_P(B) = \bar{x} + \text{Cone}(\{r^j : j \in J\}).$$

where the directions $r^j \in \mathbb{R}^n$ for $j \in J$ solve (1). ■

One can give a formula for the basic coordinates of the directions r^j for $j \in J$. The values r_i^j for $i \in B$ are given by:

$$Ar^j = A_B \begin{pmatrix} r_{i_1}^j \\ \vdots \\ r_{i_m}^j \end{pmatrix} + A_j = 0_m \iff \begin{pmatrix} r_{i_1}^j \\ \vdots \\ r_{i_m}^j \end{pmatrix} = -(A_B)^{-1} A_j = -\bar{A}_j.$$

4 The primal simplex method

The primal simplex method is an iterative method that starts with a primal feasible basis B .

It is first tested whether the information available at B allows one to conclude that (P) is unbounded, or that the associated basic solution $\bar{x}$ is optimal for (P). If this is possible, the primal simplex method terminates giving this information.

If not, the primal method attempts to identify an adjacent primal feasible basis B' whose associated basic solution has better primal objective value than $\bar{x}$. The primal simplex method is then iterated with B' as the new starting basis.

Hence it suffices to describe a single iteration of the primal simplex method. Therefore suppose B is a primal feasible basis, let $\bar{x}$ be the corresponding basic feasible solution for (P), and let $\bar{y}$ be the corresponding basic solution for (D).

The vector $\bar{s} = c - A^T \bar{y}$ denotes the slack of $\bar{y}$ in the inequalities $A^T y \leq c$. The dual slack variables are also called the *reduced costs* associated with the basis B .

Observe that the reduced costs give how much the primal objective value is decreased by moving one unit from $\bar{x}$ in the direction r^j , *i.e.*, one has:

$$c^T r^j = c_j - (c^B)^T \bar{A}_j = c_j - (c_B)^T A_B^{-1} A_j = c_j - \bar{y}^T A_j = \bar{s}_j$$

If $\bar{s} \geq 0_n$, then $\bar{y}$ is feasible for (D). Therefore, since we assumed $\bar{x}$ is basic feasible for (P), and since $\bar{x}$ and $\bar{y}$ have the same objective value in their respective problems, weak duality shows $\bar{x}$ and $\bar{y}$ are both optimal solutions. Hence the primal simplex method terminates if $\bar{s} \geq 0_n$.

Otherwise a non-basic variable $j_* \in J$ that satisfies $\bar{s}_{j_*} < 0$ is chosen (one has $\bar{s}_i = 0$ for all basic variables $i \in B$). The primal simplex method searches for a better primal feasible solution than $\bar{x}$ along the halfline that starts at $\bar{x}$ and moves in the extreme direction r^{j_*} of $\text{rec}(F_P(B))$, *i.e.*, along the halfline:

$$H^* := \{x(\theta) : \theta \geq 0\},$$

where $x(\theta) := \bar{x} + \theta \cdot r^{j_*}$.

All points on the halfline H^* (except $\bar{x}$) have better objective value than $\bar{x}$. Indeed for any point $x(\theta)$ on H^* with $\theta > 0$ one has:

$$c^T x(\theta) = c^T (\bar{x} + \theta \cdot r^{j_*}) = c^T \bar{x} + \theta \cdot \bar{s}_{j_*} < c^T \bar{x}$$

The primal simplex method tries to improve the primal objective value by moving as far as possible along the halfline H^* while preserving primal feasibility, *i.e.*, the primal simplex method computes the value:

$$\theta^* := \sup\{\theta : Ax(\theta) = b \text{ and } x(\theta) \geq 0_n\}.$$

Since $Ar^{j_*} = 0_m$ the equalities $Ax(\theta) = b$ are satisfied for all $\theta \in \mathbb{R}$. Furthermore, one has $r_k^{j_*} \in \{0, 1\}$ for all $k \in J$, and therefore $x_k(\theta) = \bar{x}_k + \theta \cdot r_k^{j_*} \geq 0$ holds for all $\theta \geq 0$ and non-basic coordinates $k \in J$. It follows that only the basic coordinates of r^{j_*} limit the size of θ . Hence:

$$\begin{aligned} \theta^* &= \sup\{\theta : \bar{x}_i + \theta \cdot r_i^{j_*} \geq 0 \text{ for all } i \in B\} \\ &= \sup\{\theta : \bar{x}_{i_k} - \theta \cdot \bar{a}_{k,j_*} \geq 0 \text{ for all } k \in \{1, \dots, m\}\} \\ &= \inf\left\{\frac{\bar{x}_{i_k}}{\bar{a}_{k,j_*}} : k \in \{1, \dots, m\} \text{ and } \bar{a}_{k,j_*} > 0\right\} \end{aligned}$$

If $\theta^* = +\infty$ (which is the case when $\bar{a}_{k,j_*} \leq 0$ holds for all $k \in B$), then $x(\theta)$ is feasible for (P) for all $\theta \geq 0$ and therefore (P) is unbounded. In this case the primal simplex method terminates.

If $\theta^* < +\infty$, then $\theta^* = \frac{\bar{x}_{i_{k_*}}}{\bar{a}_{k_*,j_*}}$ holds for some basic $i_{k_*} \in B$ that satisfies $\bar{a}_{k_*,j_*} > 0$. Since $\bar{a}_{k_*,j_*} \neq 0$, Observation 1 shows that the set:

$$B' := (B \cup \{j_*\}) \setminus \{i_{k_*}\}$$

is an adjacent basis to B .

Also, the vector $x(\theta^*)$ is the primal basic feasible solution associated with B' , i.e., one has $Ax(\theta^*) = b$ and $x_j(\theta^*) = 0$ for all $j \in N \setminus B'$. Indeed, Lemma 2 shows $Ar^{j*} = 0_m$, and therefore $Ax(\theta^*) = A\bar{x} = b$. Lemma 2 also shows $r_k^{j*} = 0$ for all $k \in J \setminus \{j_*\}$, which implies $x_k(\theta^*) = \bar{x}_k = 0$ for all $k \in J \setminus \{j_*\}$. Finally:

$$\begin{aligned} x_{i_{k_*}}(\theta^*) &= \bar{x}_{i_{k_*}} - \theta^* \cdot \bar{a}_{k_*, j_*} && (\text{ since } r_{i_{k_*}}^{j_*} = -\bar{a}_{k_*, j_*}) \\ &= \bar{x}_{i_{k_*}} - \left(\frac{\bar{x}_{i_{k_*}}}{\bar{a}_{k_*, j_*}} \right) \cdot \bar{a}_{k_*, j_*} = 0. && (\text{ from the formula for } \theta^*) \end{aligned}$$

We conclude that B' is a primal feasible basis with $x(\theta^*)$ as the associated basic solution, and that $x(\theta^*)$ has better primal objective value than $\bar{x}$ when $\theta^* > 0$.

Observation 2 (*An adjacent basis*)

The basis B' is primal feasible and adjacent to B , and $x(\theta^)$ is the associated basic solution. Also, $x(\theta^*)$ has better primal objective value than $\bar{x}$ if $\theta^* > 0$. ■*

If it has not been discovered that (P) is unbounded, or that $\bar{x}$ is optimal for (P), the primal simplex method performs another iteration with B' as the new starting basis. The procedure is continued until either an optimal solution has been found, or it has been discovered that (P) is unbounded.

There is no guarantee of improvement in every iteration, i.e., one may have $\theta^* = 0$. However, this turns out not to be a big problem in practice.

5 Example

We consider an example with $n = 4$ variables and $m = 2$ equality constraints. Hence $N = \{1, 2, 3, 4\}$. The set F_P of feasible solution is:

$$F_P = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 : \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \text{ and } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \geq 0_4 \right\}$$

with objective function $c^T = (2, 0, 0, 0)$. In the example we therefore have:

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 3 & 4 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

The set $B = \{1, 2\}$ is a basis, since the vectors $A_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ are linearly independent. We choose $A_B = [A_1, A_2]$ as the basis matrix, and therefore $i_1 = 1$ and $i_2 = 2$. The basis inverse of A_B is given by:

$$A_B^{-1} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}.$$

and the columns in the updated matrix $\bar{A} = A_B^{-1}A$ are given by:

$$\bar{A}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \bar{A}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \bar{A}_3 = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix} \text{ and } \bar{A}_4 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

The primal basic solution $\bar{x} \in \mathbb{R}^4$ associated with B is the solution to:

$$x_3 = x_4 = 0 \text{ and } Ax = b \iff x_3 = x_4 = 0 \text{ and } A_B \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

and therefore $\bar{x}_3 = \bar{x}_4 = 0$ and $\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix} = A_B^{-1} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. We conclude that the primal basic solution $\bar{x} \in \mathbb{R}^4$ associated with B is:

$$\bar{x}^T = (1 \quad 1 \quad 0 \quad 0).$$

which is feasible for (P) since $\bar{x} \geq 0_4$ and $A\bar{x} = b$.

One has $c^B = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, so the dual basic solution is:

$$\bar{y} = (c^B)^T A_B^{-1} = (0 \quad 1),$$

which is not feasible for (D) since the reduced costs/dual slacks given by $\bar{s}_3 = c_3 - A_3^T \bar{y} = -3 < 0$ and $\bar{s}_4 = c_4 - A_4^T \bar{y} = -4 < 0$ are negative.

The non-basic variables are $J = \{3, 4\}$. The non-basic coordinates of r^3 and r^4 are given by $r_3^3 = r_4^3 = 1$, and $r_3^4 = r_4^4 = 0$. The basic coordinates of r^3 and r^4 are $\begin{pmatrix} r_1^3 \\ r_2^3 \end{pmatrix} = -\bar{A}_3 = \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix}$ and $\begin{pmatrix} r_1^4 \\ r_2^4 \end{pmatrix} = -\bar{A}_4 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, and therefore:

$$r^3 = \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} \text{ and } r^4 = \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Observe that the directions r^3 and r^4 are indeed solutions to the homogenous system $Ax = 0_2$, *i.e.*, one has $Ar^3 = 0_2$ and $Ar^4 = 0_2$ (Lemma 2).

Recall that the objective function in our example is given by $c^T = (2, 0, 0, 0)$. The reduced costs $\bar{s}_3$ and $\bar{s}_4$ associated with the non-basic variables $J = \{3, 4\}$ can therefore also be calculated as:

$$\bar{s}_3 = c^T r^3 = c^T \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} = -3 \quad \text{and} \quad \bar{s}_4 = c^T r^4 = c^T \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix} = -4.$$

Hence moving from $\bar{x}$ along any of the halflines $\{\bar{x} + \theta r^3 : \theta \geq 0\}$ and $\{\bar{x} + \theta r^4 : \theta \geq 0\}$ will lead to a point which has a better primal objective function value than $\bar{x}$.

We consider the halfline $\{\bar{x} + \theta r^3 : \theta \geq 0\}$ in our example. In order for it to be both feasible and profitable to move from $\bar{x}$ to a point $\bar{x} + \theta r^3$, one needs θ to satisfy:

$$\bar{x} + \theta r^3 \geq 0_4 \text{ and } \theta > 0 \iff \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \theta \begin{pmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} \geq 0_4 \text{ and } \theta > 0 \iff 0 < \theta \leq \frac{2}{3}.$$

Hence a better solution to (P) than $\bar{x}$ is the point:

$$x(\theta^*) = \bar{x} + \theta^* r^3 = \bar{x} + \frac{2}{3} r^3 = \begin{pmatrix} 0 \\ \frac{4}{3} \\ \frac{2}{3} \\ 0 \end{pmatrix}$$

with objective function value $c^T(\bar{x} + \frac{2}{3} r^3) = c^T \bar{x} + \frac{2}{3} \bar{c}_3 = 0 < c^T \bar{x} = 2$.

The above operation corresponds to taking the variable x_3 into the basis and taking the variable x_1 out of the basis:

$$B' := (B \setminus \{1\}) \cup \{3\} = \{2, 3\},$$

and the basic feasible solution corresponding to B' is:

$$x(\theta^*) = \begin{pmatrix} 0 \\ \frac{4}{3} \\ \frac{2}{3} \\ 0 \end{pmatrix}$$

■